YIRAGE (YPK) — SYSTEM ARCHITECTURE

RuntimeFusion × Multi-Backend · source: docs/yirage-architecture.drawio + include/backend/*.h · github.com/chenxingqiang/YiRage

INFERENCE ENGINE (scheduling + KV owner)

- **vLLM** — continuous batch · PagedAttention
- **SGLang** — RadixAttention · extend meta
- **meta** — block_table · seq_lens · SM budget

decode

RUNTIMEFUSION yirage.serving · RF.step(inputs, meta) selects 0..N capsules / decode step

- **RF.step()** — yirage.serving
- capsule
- **FusionCapsule MLP** — gate_up + down yirage.core
- **FusionCapsule seg** — gate_up → down pipeline
- **skip / fallback** — engine native ops

plan

SEARCH & PLAN (offline / warm-up)

- **superoptimize(backend)** — KN/TB graph search
- **FusionPlan / μGraph** — profile + fingerprint verify
- **MuGraph cache** — shape-bucket restore
- **Ray coordinator** — distributed search optional
- **serving tiers** — seed_verify · full_tb · full_tb_ray

PYTHON API & BACKEND MANAGER (C++)

- **PersistentKernel** — python/yirage
- **KNGraph / TBGraph** — transpiler → kernels
- **BackendRegistry** — include/backend/backend_registry.h
- **yirage.core** — Cython runtime
- **KernelExecutor factory** — search + execute same backend

BACKEND IMPLEMENTATIONS include/backend/*.h + include/kernel/{cuda,maca,cpu,mps,ascend}/

- **CUDA** — cuda_backend.h warp 32 nvcc
- **MACA** — maca_backend.h warp 64 mxcc
- **ROCm** — rocm_backend.h AMD GPU
- **MPS** — mps_backend.h Apple GPU
- **Ascend** — ascend_backend.h CANN
- **CPU** — cpu_backend.h OpenMP/MKL
- **Triton** — triton_backend.h DSL
- **NKI** — nki_backend.h SBUF/DMA
- **cuDNN** — cudnn_backend.h
- **MKL** — mkl_backend.h
- **TPU** — tpu_backend.h
- **XPU** — xpu_backend.h
- **MLIR** — mlir_backend.h CPU JIT
- **FPGA** — fpga_backend.h experimental

HARDWARE TARGETS

- **NVIDIA GPU** — Ampere / Hopper
- **MetaX C500/C550** — MACA
- **x86 / ARM CPU** — Intel / AMD / Apple
- **Apple Silicon** — M1–M4
- **Huawei Ascend** — 910 / 910B / 310P
- **AWS Neuron / Intel XPU** — NKI

Engine owns scheduling & KV; RF owns fusion identity per step · BackendRegistry search+execute same backend